

NAME : P.KRISHNAPRIYA

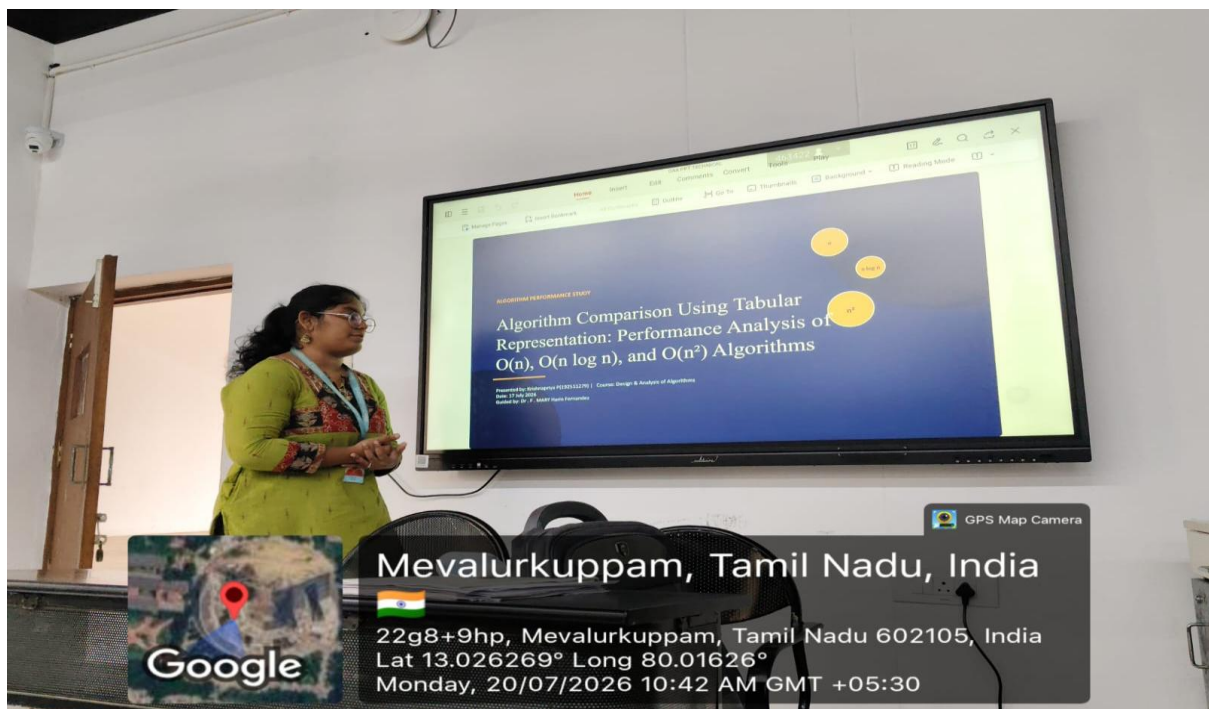
COURSE CODE : CSA0613

REG NO : 192511279

**COURSE NAME :Design and
Analysis of Algorithms**

C01-ASSESSMENT TOOL 03 -TECHNICAL PRESENTATION

TITLE : "Algorithm Comparison Using Tabular Representation: Performance Analysis of $O(n)$, $O(n \log n)$, and $O(n^2)$ Algorithms,"



PROJECT DESCRIPTION

This project, "Algorithm Comparison Using Tabular Representation: Performance Analysis of $O(n)$, $O(n \log n)$, and $O(n^2)$ Algorithms," focuses on analyzing and comparing the efficiency of three common algorithm time complexities using a tabular representation. The study evaluates how each algorithm performs as the input size increases by measuring the number of operations and execution trends. Sample algorithms representing linear, linearithmic, and quadratic complexities are considered to demonstrate their behavior on different dataset sizes. The collected data is organized into tables to clearly highlight the differences in performance, scalability, and computational efficiency. Based on the analysis, the project identifies the most suitable algorithm for handling small and large datasets, helping learners understand the importance of time complexity in selecting efficient algorithms for real-world applications.